

Rhea Sudheer

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Summary

Software engineer with an M.S. in Computer Science from Texas A&M and experience developing **production backend systems, data-intensive web applications, REST APIs, and relational data models**. Background spans **role-based access, data validation, healthcare software, distributed services, and AI/LLM systems**, with an emphasis on **reliable system design, clear component boundaries, reproducible engineering workflows, and maintainable software** across application, database, data, and user-facing layers.

Education

Texas A&M University — M.S. Computer Science, GPA: **3.83/4.00**

Aug 2024 – May 2026

PES University — B.Tech. Computer Science, GPA: **3.88/4**

Dec 2020 – May 2024

Technical Skills

Languages / Backend: Python, PHP, SQL, JavaScript, Java, C++, Laravel, Livewire, Flask, REST APIs, MySQL, HTML/CSS, Bootstrap
Systems / Data: Docker, Kubernetes, Jenkins, Git, CI/CD, Linux, relational data modeling, ETL, AWS, GCP

AI / ML: LLMs, RAG, Model Context Protocol (MCP), LangChain, FAISS, BM25, Sentence-Transformers, Hugging Face Transformers, PyTorch, TensorFlow, Scikit-learn

Experience

Public Policy Research Institute, Texas A&M University

College Station, TX

Software Applications Developer I (*Full-time*)

Sep 2026 – Present

Student Research Assistant

Feb 2025 – Sep 2026

- Engineer and maintain a production **PHP/Laravel, Livewire, MySQL, and REST API** research platform supporting data collection, validation, multi-stage review workflows, dashboards, and reporting across **100,000+ public records**.
- Build reusable application components for **dynamic forms, permissions, validation, and role-based workflows**, enforcing application rules across backend logic, persisted state, and user-facing behavior while supporting different review and administrative roles.
- Design relational schemas, reusable backend services, SQL transformation logic, and API-backed **ETL pipelines**; normalize heterogeneous source data and automate detection of malformed, incomplete, duplicate, and inconsistent records before downstream processing.
- Develop metadata-driven workflows and reusable application logic to support evolving research requirements without duplicating feature-specific implementations across forms, reviews, and reporting interfaces.
- Own **end-to-end feature development** across application, database, ingestion, review, and reporting layers; translate evolving requirements into production features, strengthen validation and regression protection, and troubleshoot production workflows directly with users.

IBM | Tech Mentee, Healthcare Machine Learning

Dec 2022 – Mar 2023

- Built an end-to-end **Type-I diabetes prediction pipeline** over heterogeneous multi-omic healthcare data, spanning validation, preprocessing, feature engineering, model selection, cross-validation, evaluation, inference, and interpretability.
- Standardized heterogeneous biological features into a reproducible modeling matrix and compared decision-tree and ensemble-learning approaches through controlled validation, achieving **97.7% reported prediction accuracy**.
- Built reproducible preprocessing, training, validation, and inference stages to maintain consistent transformations across experimentation; applied interpretability-oriented analysis to investigate influential biological variables and healthcare relationships.

Thales | Artificial Intelligence Intern

Jun 2022 – Aug 2022

- Developed a real-time **MobileNetV2 + SSD** pedestrian-detection pipeline, profiling inference latency, detection quality, model size, and false/missed detections; integrated **SHAP, LIME, and saliency mapping** for explainability, debugging, and systematic failure analysis.

Projects

E-commerce Microservices Application | Docker, Kubernetes, Jenkins, REST APIs

- Built a **microservices-based application** separating user management, product management, and cart/order functionality into distinct services with independent application responsibilities and a web-based frontend.
- Containerized services with **Docker**, defined **Kubernetes** deployment configuration, and implemented a **Jenkins CI pipeline** to support repeatable build and deployment workflows.

Report Genie — Healthcare Document Intelligence | Python, Flask, Gemini, FAISS, BM25

Jan 2025

- Engineered an end-to-end **healthcare application** spanning medical-PDF ingestion, structured clinical-value extraction, evidence retrieval, LLM inference, interactive patient Q&A, and a Flask-based user-facing application.
- Built a **hybrid sparse+dense retrieval pipeline** combining BM25, Sentence-Transformer embeddings, and FAISS vector search to retrieve medically relevant evidence for grounded report explanations and question answering.
- Modularized **document parsing, structured extraction, retrieval, prompt construction, and generation**, enabling component substitution, targeted debugging, and failure isolation across the end-to-end workflow.

Hybrid Self-RAG with External Knowledge Validation | GPT-4, LangChain, FAISS, Transformers

May 2025

- Designed a multi-stage **LLM validation architecture** combining atomic claim extraction, external evidence retrieval, semantic grounding evaluation, and conditional retrieval-augmented regeneration for insufficiently supported outputs.
- Embedded generated claims and retrieved evidence with Sentence-Transformers and applied a **FactScore-based threshold** to route insufficiently grounded answers through FAISS fallback retrieval and evidence-conditioned regeneration.
- Improved factual accuracy from **81% → 89%**, reduced hallucination rate from **19% → 11%**, and increased average FactScore from **0.65 → 0.87** versus a Self-RAG-only baseline.

Publications

Revolutionizing Healthcare: A Review Unveiling the Transformative Power of Digital Twins

Enhancing Robustness in Audio Visual Speech Recognition: Preprocessing Approach with Transformer and CTC Loss

Leukaemia: A Comparative Analysis of Deep Learning Models Using ALL Dataset